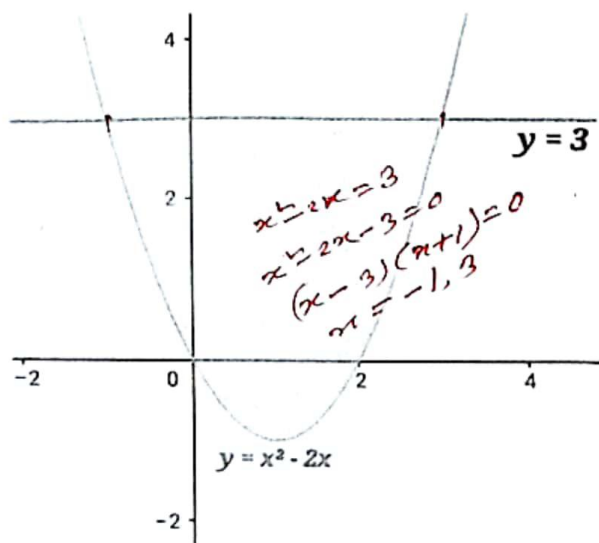
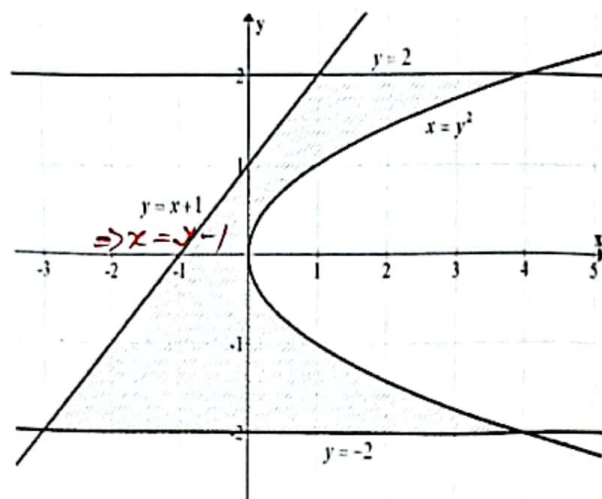


Math-1151 Class Test 04 Section A

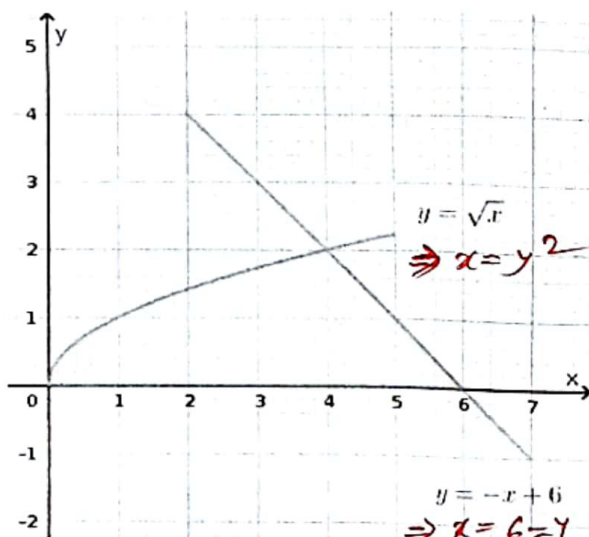
1. Find the area of the following shaded regions. $[2.5 \times 4 = 10]$



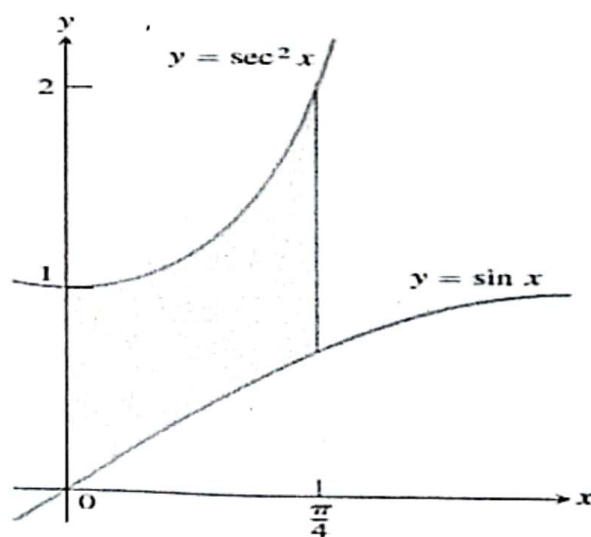
(a)



(b)



(c)



(d)

$$a. \quad A = \int_{-1}^3 [3 - (x^2 - 2x)] dx = 3x - \frac{x^3}{3} + x^2 \Big|_{-1}^3 = (9 - 9 + 9) - (-3 + \frac{1}{3} + 1) = \frac{32}{3}$$

$$b. \quad A = \int_{-2}^1 [y^2 - (y - 1)] dy = \frac{y^3}{3} - \frac{y^2}{2} + y \Big|_{-2}^1 = (\frac{8}{3} - 2 + 2) - (-\frac{8}{3} - 2 - 2) = \frac{28}{3}$$

$$c. \quad A = \int_0^2 [(6 - y) - y^2] dy = 6y - \frac{y^2}{2} - \frac{y^3}{3} \Big|_0^2 = 12 - 2 - \frac{8}{3} = \frac{22}{3}$$

$$d. \quad A = \int_0^{\pi/4} (\sec^2 x - \sin x) dx = \tan x + \cos x \Big|_0^{\pi/4} = 1 + \frac{1}{\sqrt{2}} - 0 - 1 = \frac{1}{\sqrt{2}}$$